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Department of CSE

B.Tech. CSE Trimester – 5

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Cloud Infrastructure And Services

25CS2206E

AWS Practical: MFA, Linux Instance and S3 Bucket

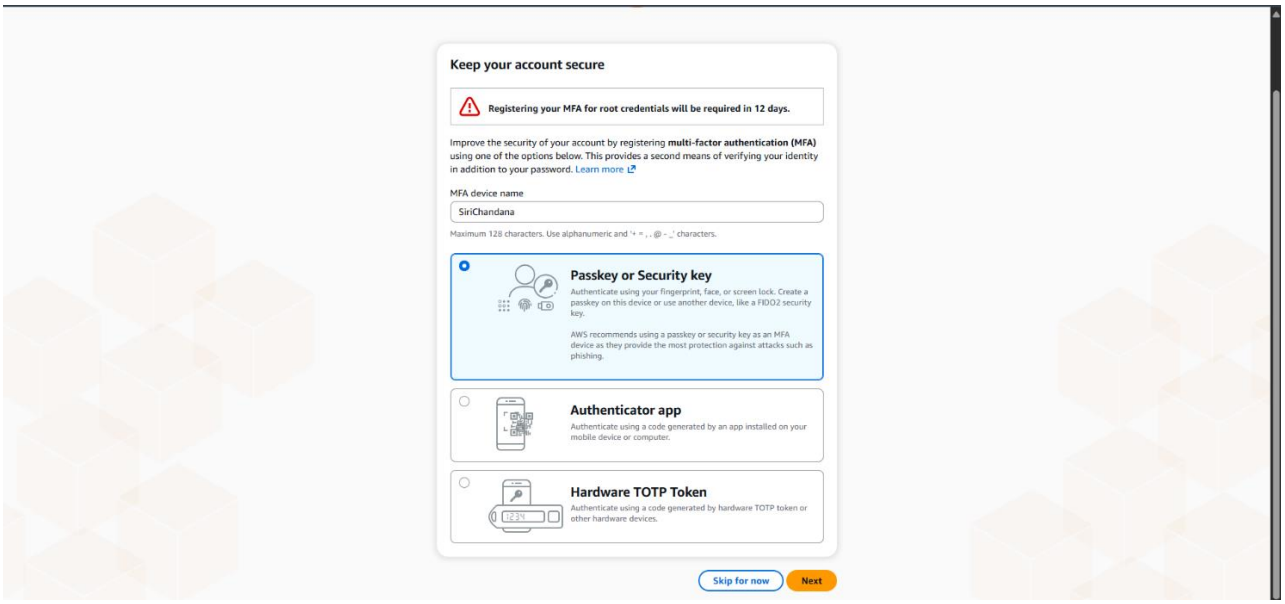
Aim

To enable Multi-Factor Authentication (MFA), create and connect to a Linux EC2 instance, and create an Amazon S3 bucket to store objects.

Introduction

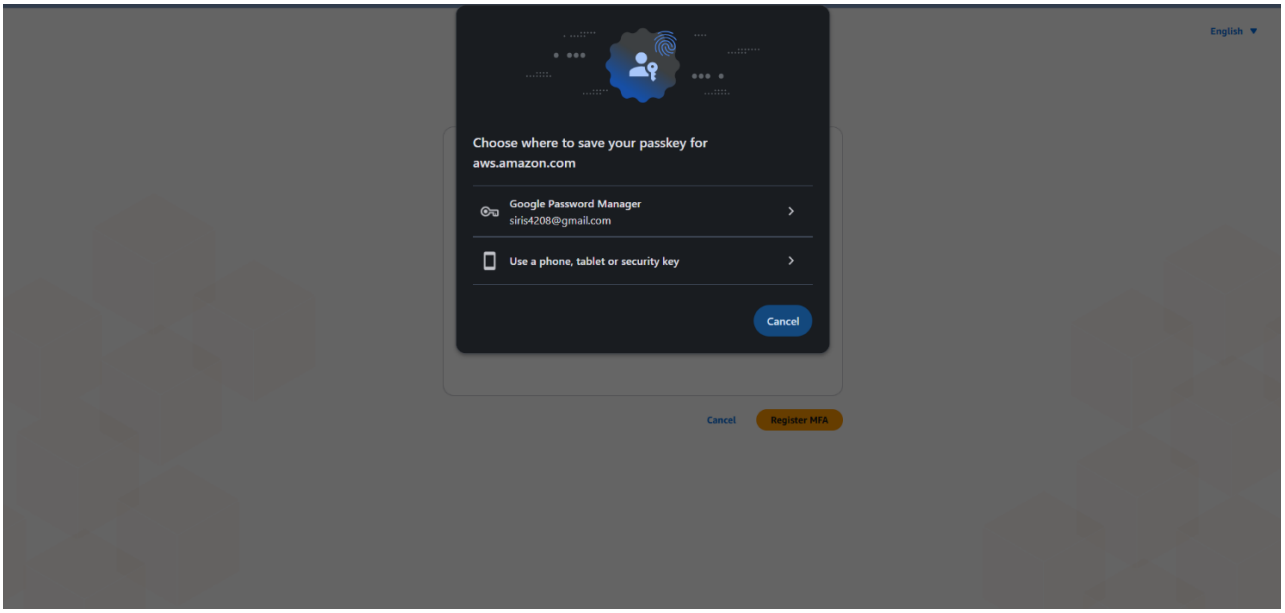
AWS provides cloud services to secure an account, run virtual computers and store files. MFA adds an extra security check, EC2 provides virtual machines, and S3 stores files in the cloud.

a) Enabling Multi-Factor Authentication (MFA)



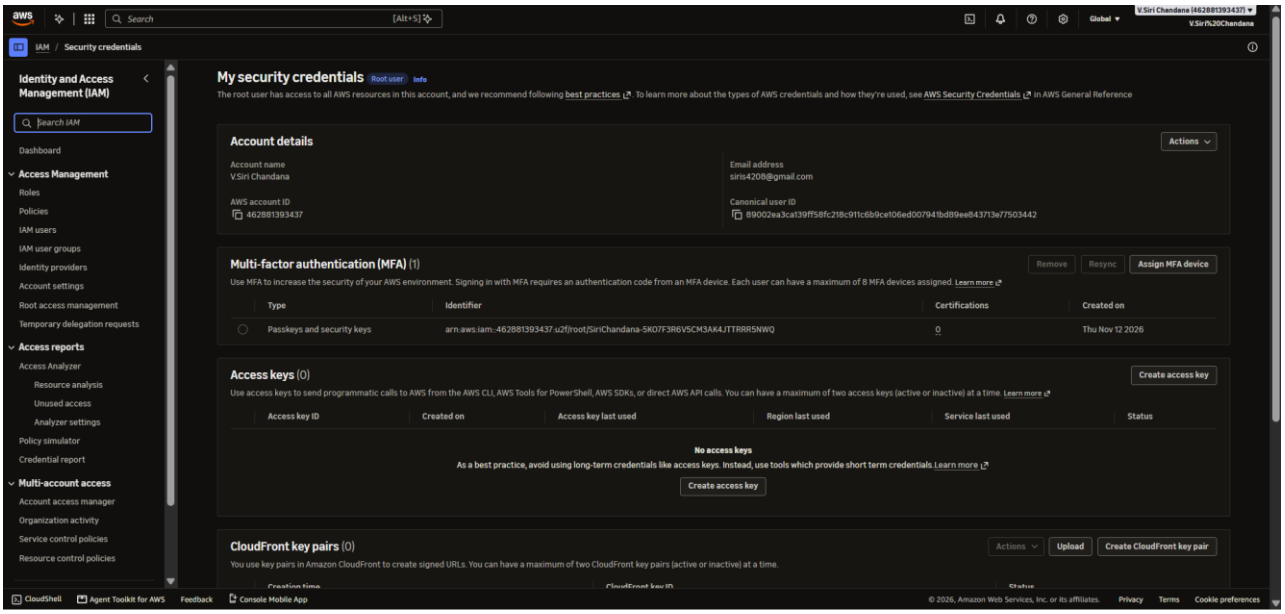
Step 1: Select an MFA method

Description: We are choosing a passkey or security key as an extra method to protect the AWS account.



Step 2: Save the passkey

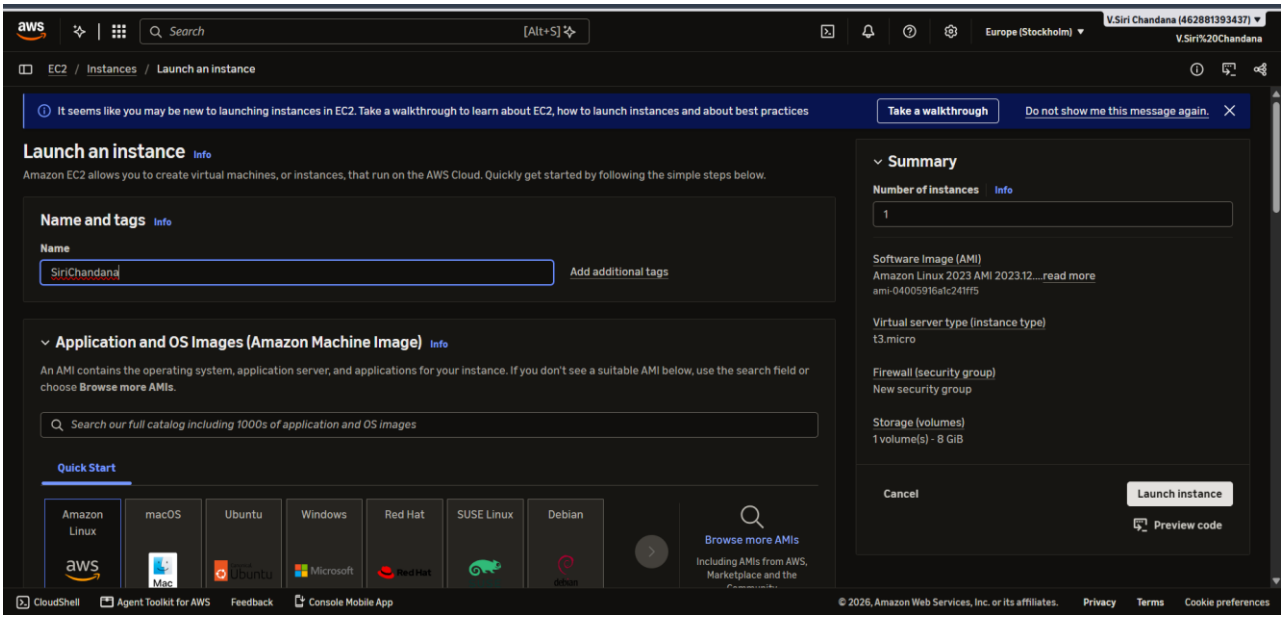
Description: We are choosing where to save the passkey so it can be used to verify our identity.



Step 3: Verify MFA registration

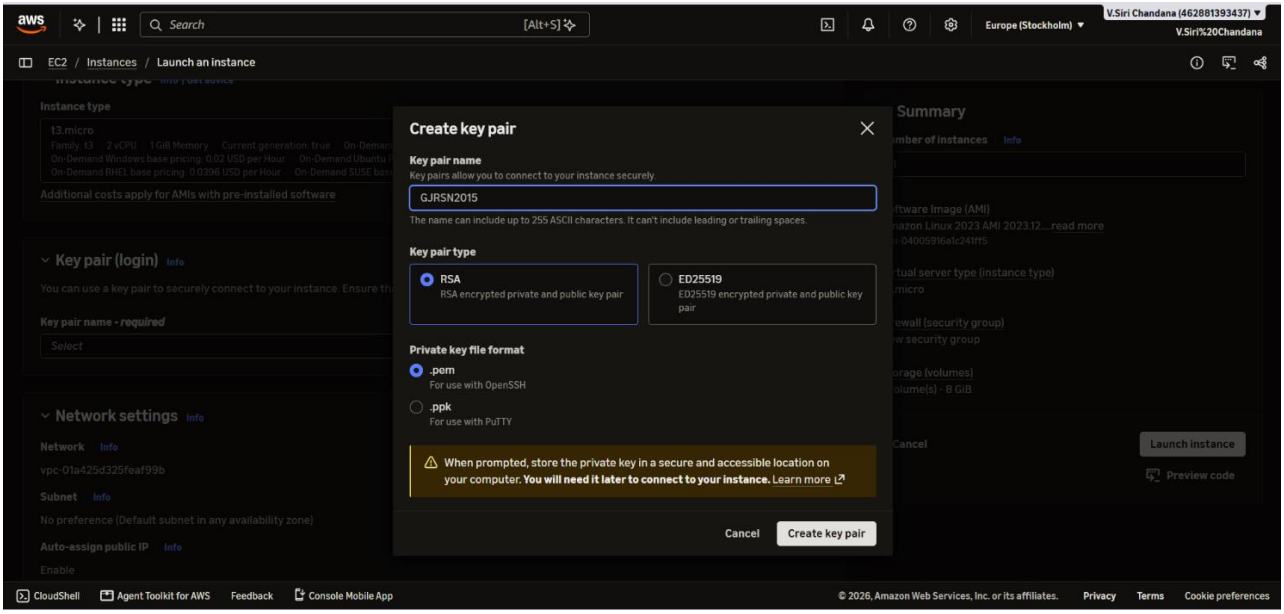
Description: We are checking the AWS security credentials page to confirm that the MFA device is registered.

b) Creating the First Linux Instance



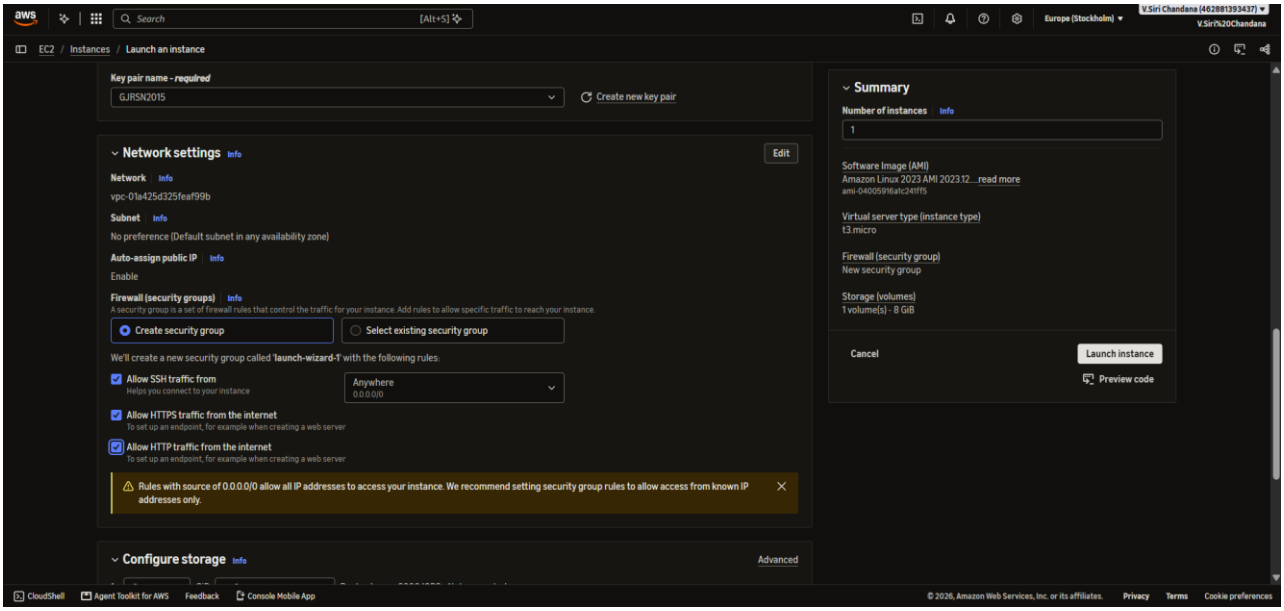
Step 1: Open the EC2 launch page

Description: We are entering the instance name and selecting an Amazon Linux image to create a Linux virtual machine.



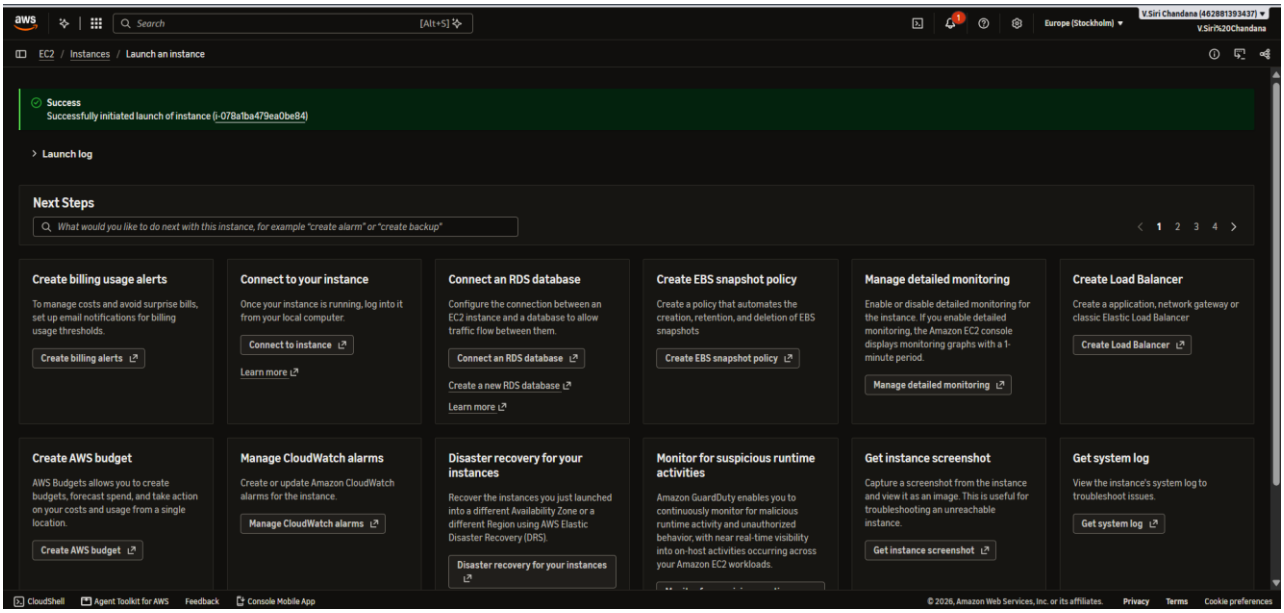
Step 2: Create a key pair

Description: We are creating a key pair that will be used to securely connect to the Linux instance.



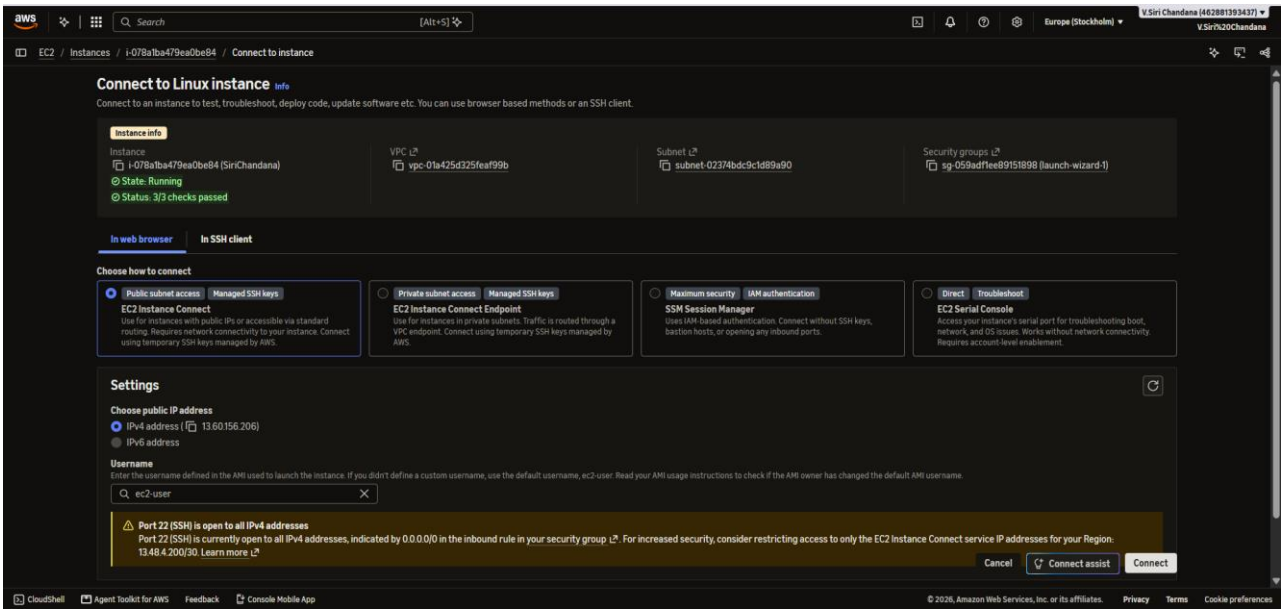
Step 3: Configure network and storage

Description: We are checking the network, security group and storage settings for the Linux instance.



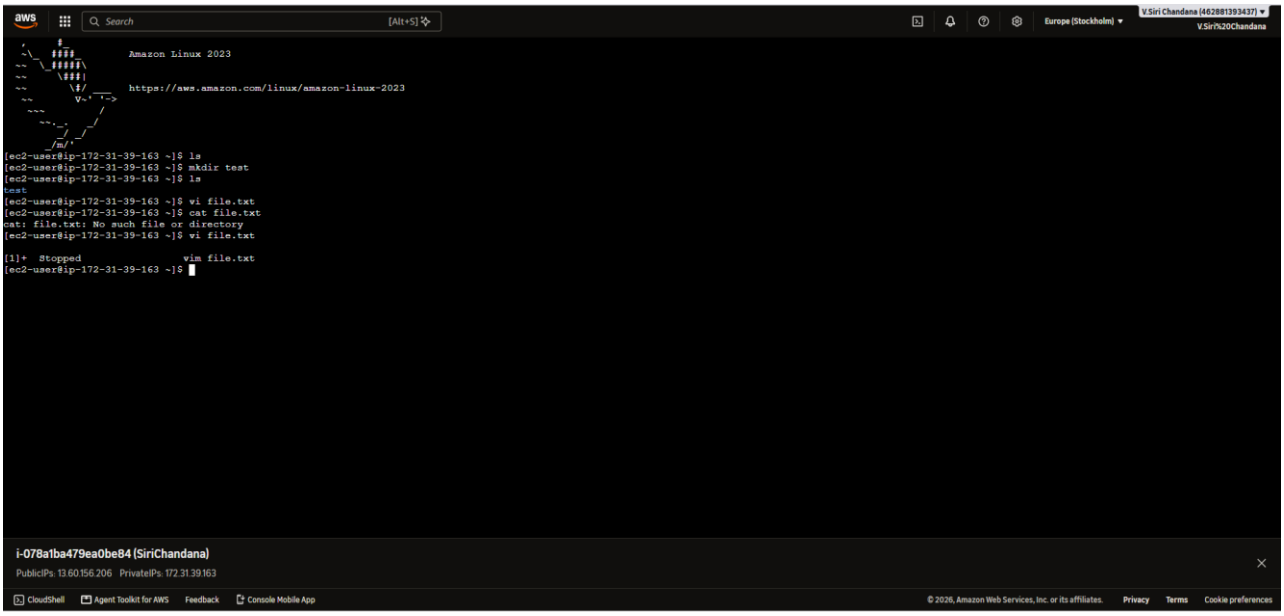
Step 4: Launch the instance

Description: We are launching the configured Linux instance and checking the success message.



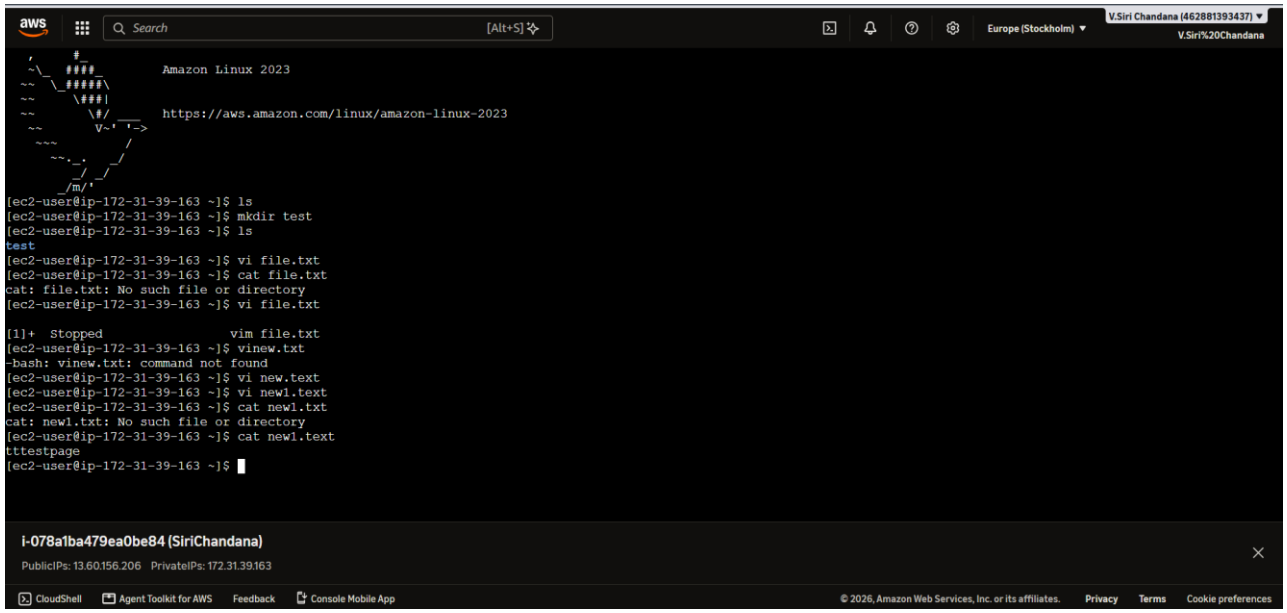
Step 5: Open the connection page

Description: We are opening the connection options for the running Linux instance.



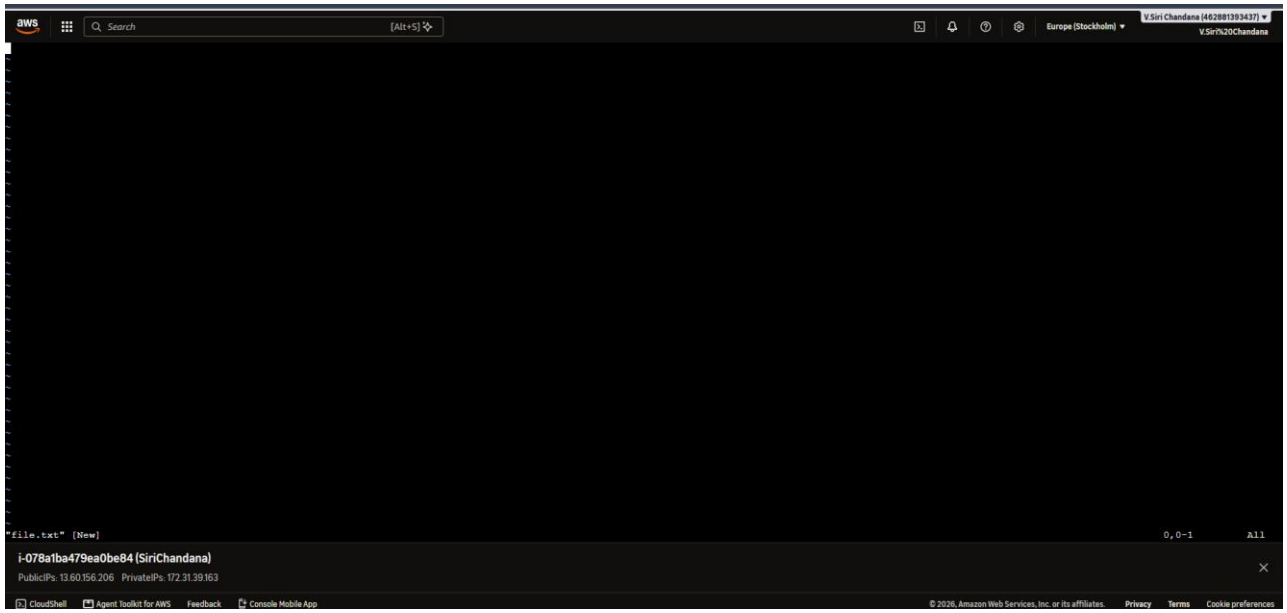
Step 6: Run Linux commands

Description: We are using the Linux terminal to list files, create a folder and work with a text file.



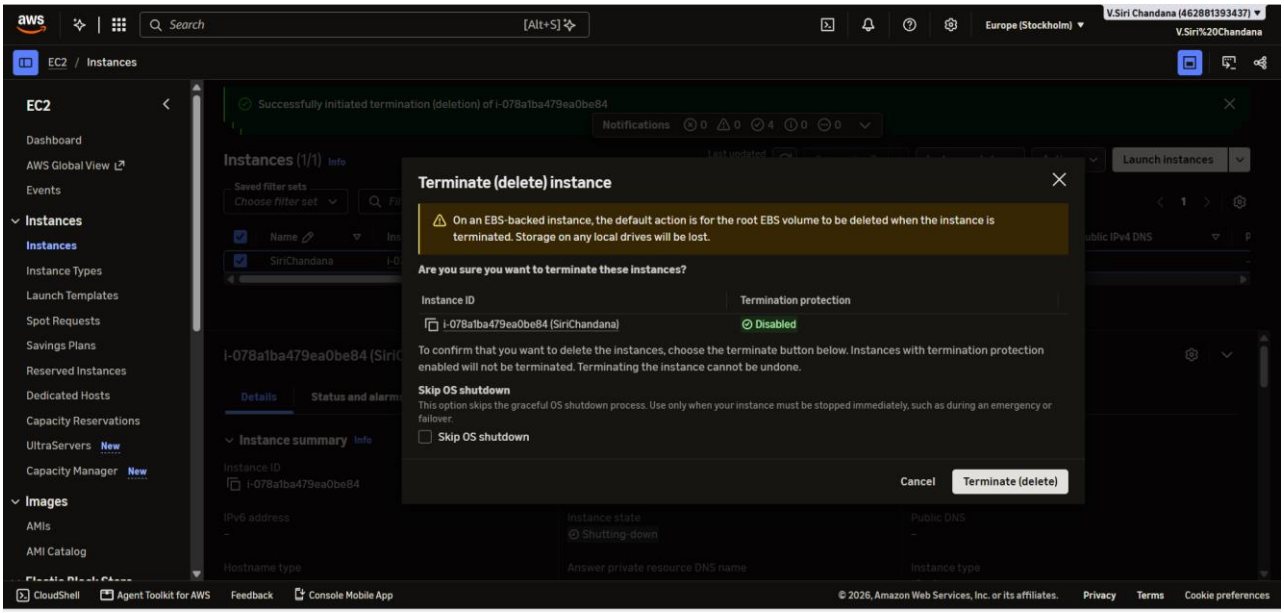
Step 7: Practice more Linux commands

Description: We are practising file editing and viewing commands in the Linux terminal.



Step 8: Open the text editor

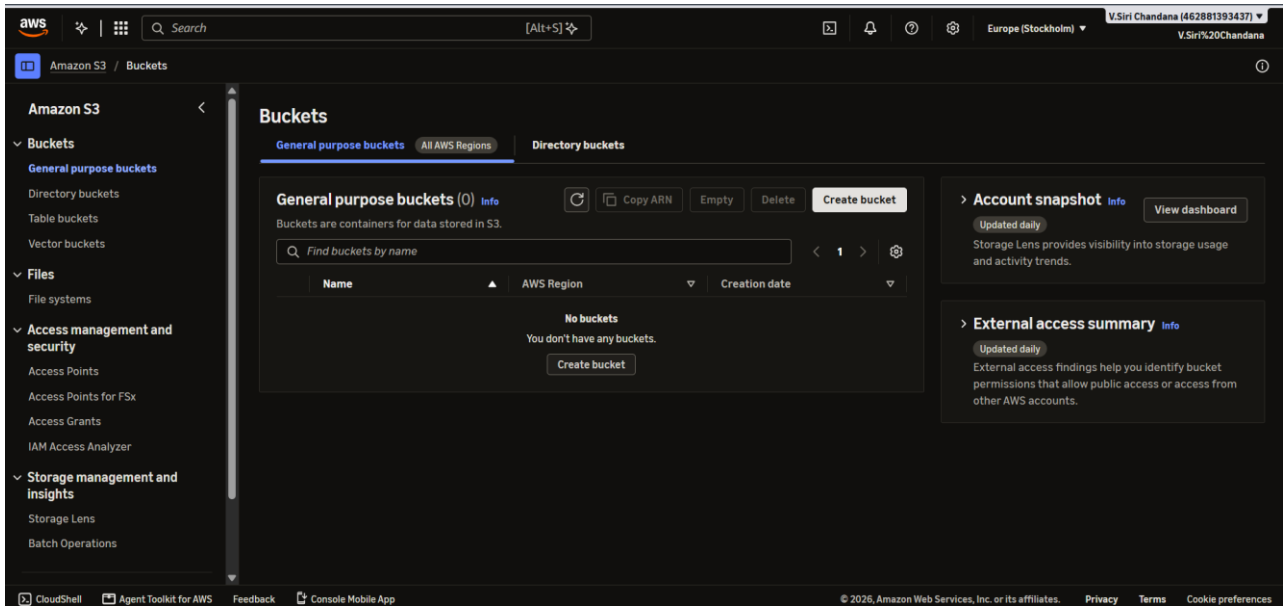
Description: We are opening the Vim editor to create or edit a text file in Linux.



Step 9: Terminate the instance

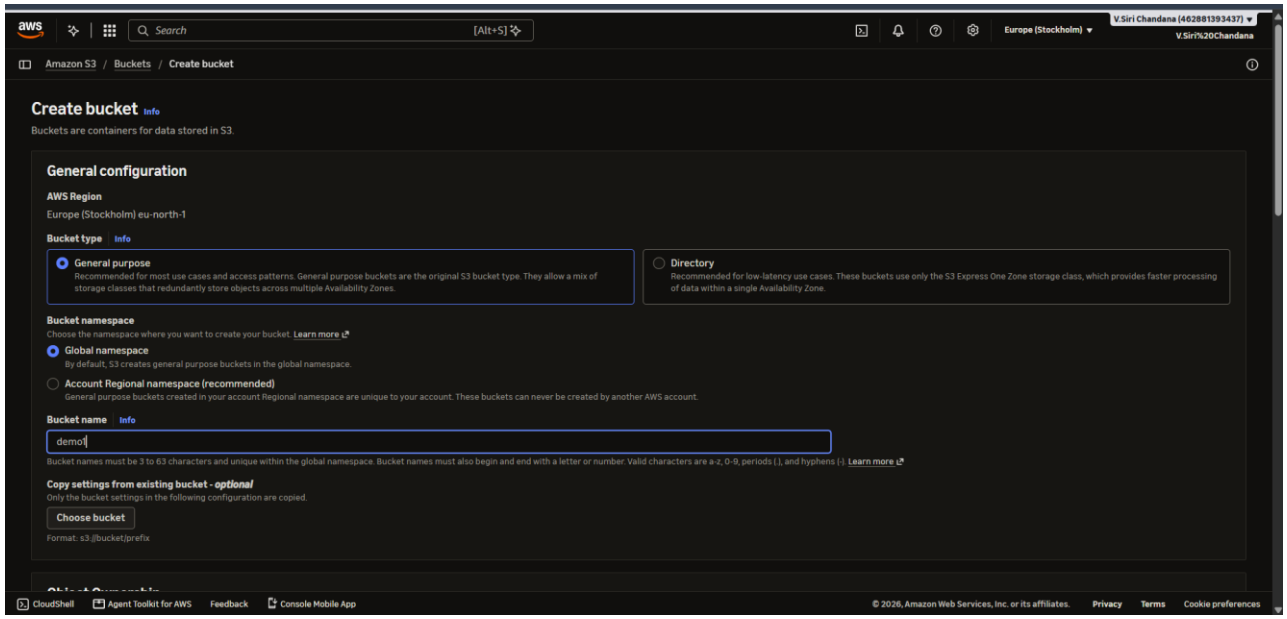
Description: We are confirming the termination of the EC2 instance to stop and delete the virtual machine.

c) S3 Bucket Creation and Uploading Objects



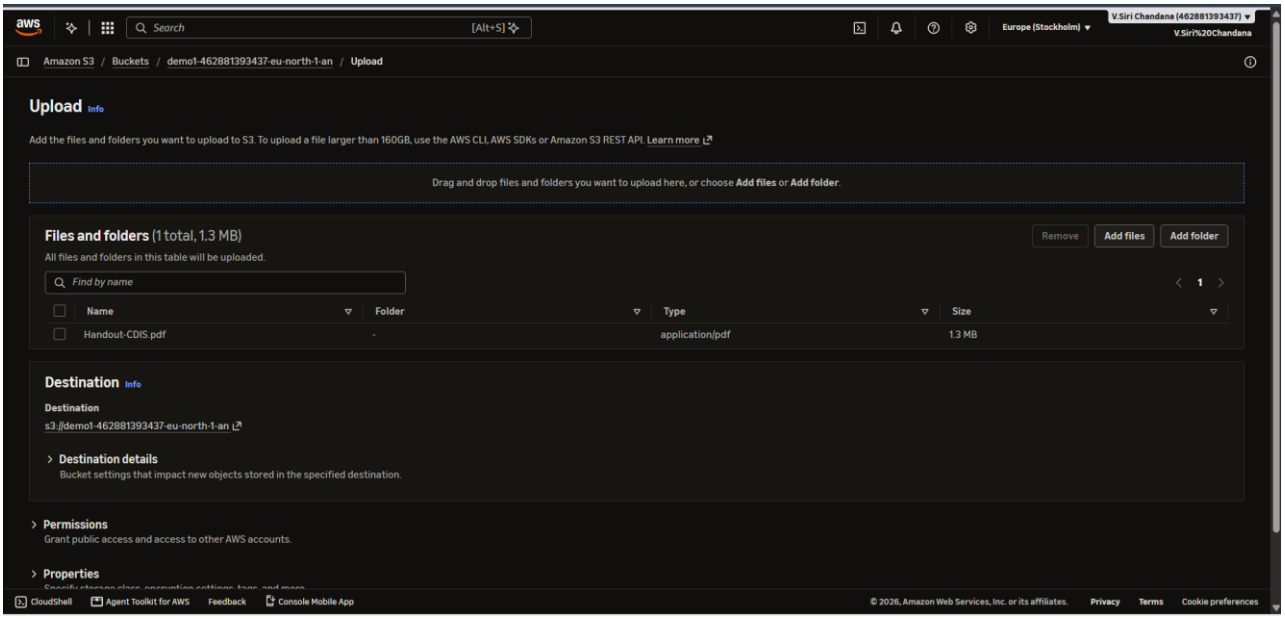
Step 1: Open the S3 dashboard

Description: We are opening Amazon S3 to create a bucket for storing files in the cloud.



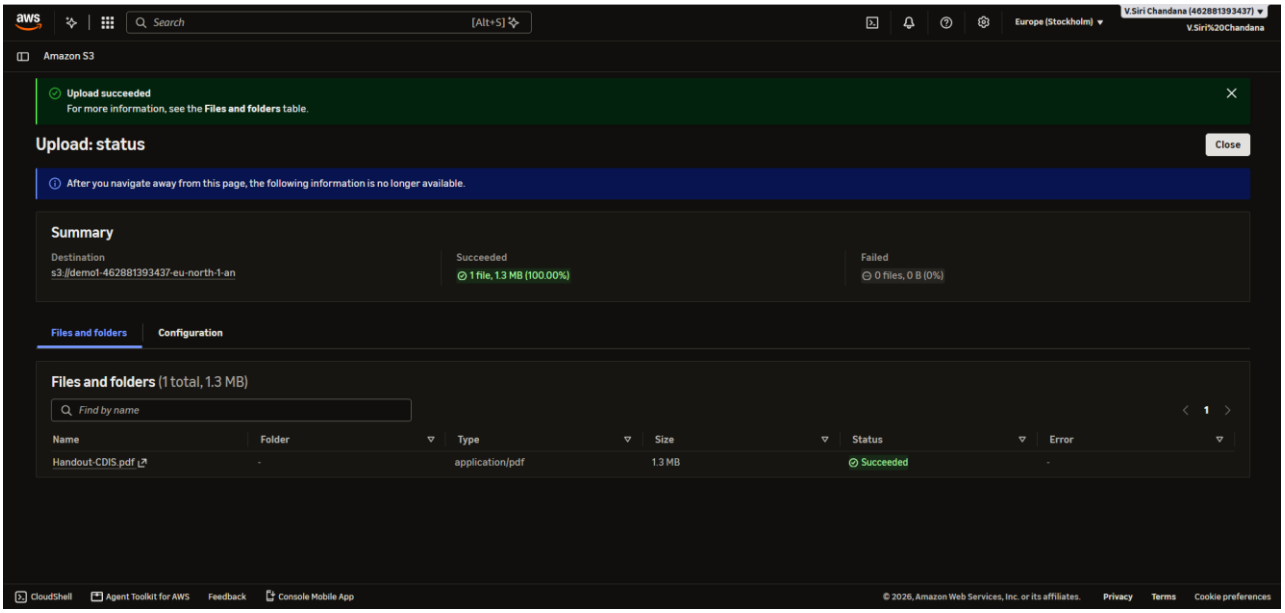
Step 2: Enter bucket details

Description: We are selecting the bucket type, choosing the AWS Region and entering a unique bucket name.



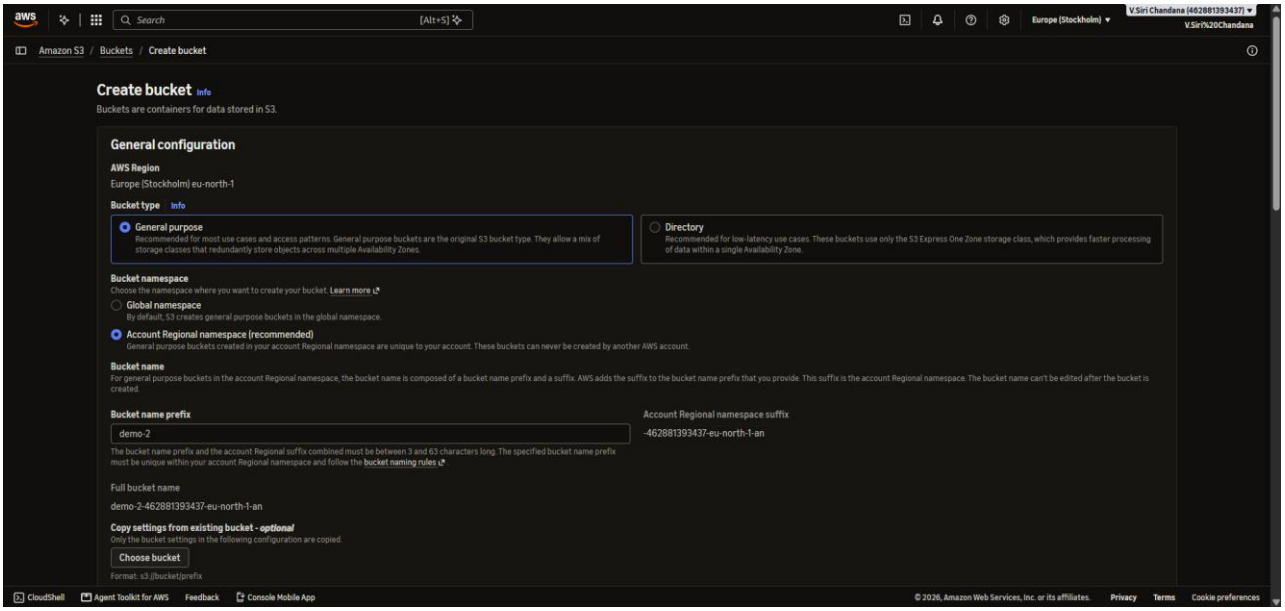
Step 3: Select a file to upload

Description: We are selecting a PDF file from the computer to upload into the S3 bucket.



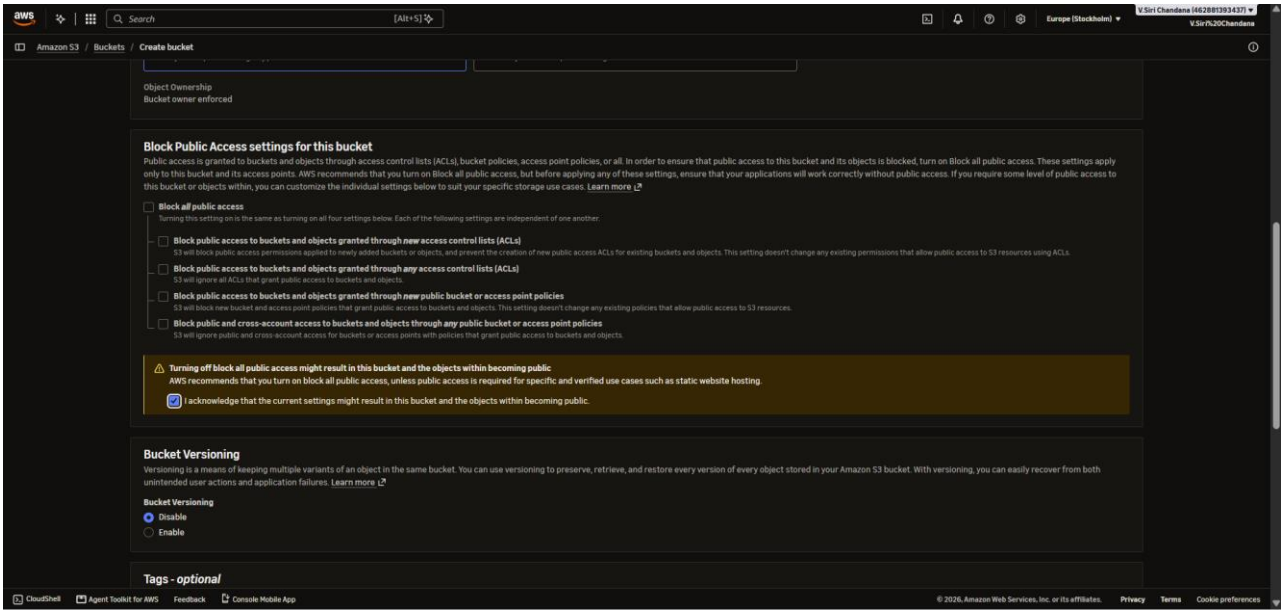
Step 4: Check upload status

Description: We are checking the upload result to confirm that the PDF file was uploaded successfully.



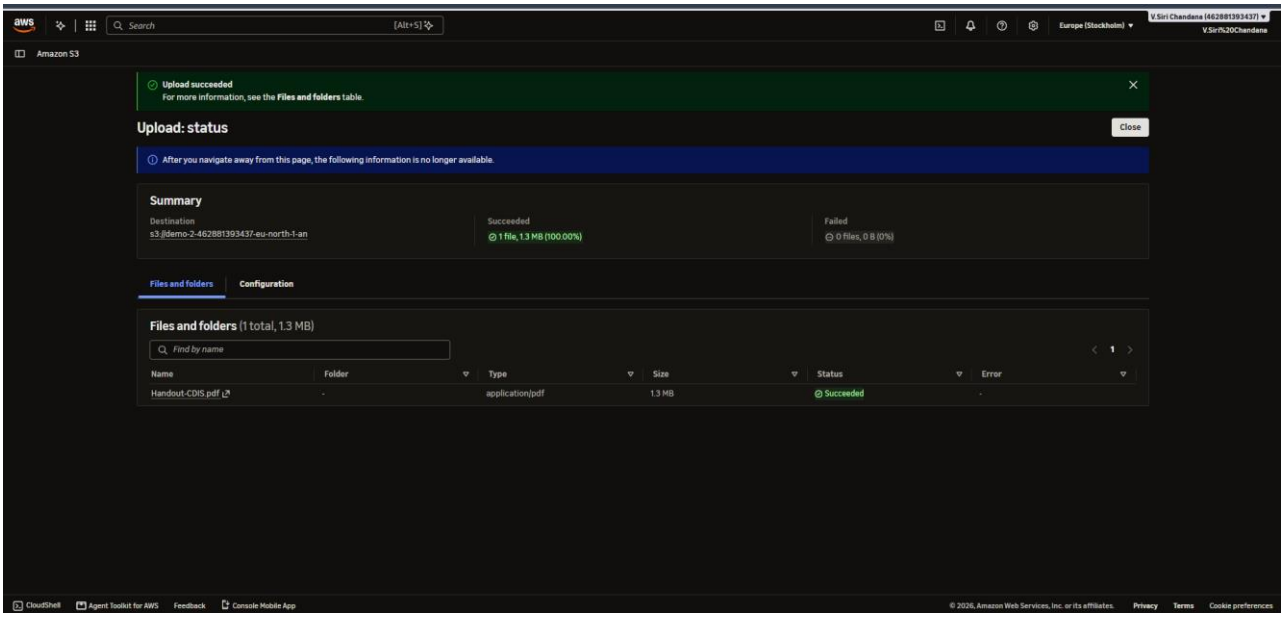
Step 5: Enter details for another bucket

Description: We are entering the name and settings for another S3 bucket.



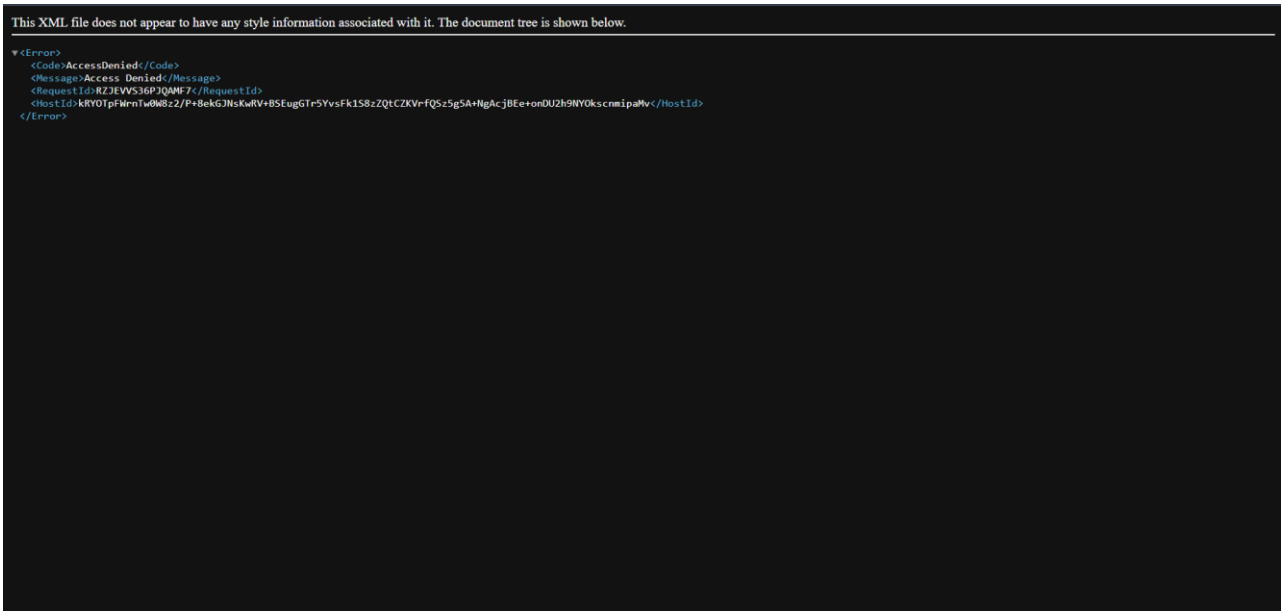
Step 6: Review bucket access settings

Description: We are reviewing the public access and versioning settings before creating the bucket.



Step 7: Verify the uploaded object

Description: We are checking the upload status and confirming that the selected PDF file is stored in the bucket.



Step 8: Check object access

Description: We are checking the object link; access is denied because the object is not publicly accessible.

Result

MFA was registered for the AWS account, a Linux EC2 instance was launched and accessed, and S3 buckets were created with a PDF object uploaded successfully.

Conclusion

We learned how to add extra security to an AWS account using MFA, create and use a Linux virtual machine through EC2, and store files using Amazon S3.